
Knowledge Discovery & Data Mining

— Syllabus —

Instructor: Jiaxin Du

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Contact Information

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- ▶ Instructor: Dr. Jiaxin Du (Dr. Du)
- ▶ LinkedIn: <https://www.linkedin.com/in/jiaxin-du-a3861134/>
- ▶ Google Scholar: <https://scholar.google.com/citations?user=LvfBANYAAAAJ&hl=en&oi=ao>
- ▶ E-mail: dujia@gvsu.edu
- ▶ Class location: DeVos Cntr Interprofess Health 505 or zoom
- ▶ Zoom information:
 - ▶ <https://gvsu-edu.zoom.us/j/98277673804?pwd=OQCFCqtyg74SSbE5Msabjp6eT5bn44.1>
 - ▶ Meeting ID: 982 7767 3804
 - ▶ Passcode: 671980
 - ▶ Online asynchronous students will see the video after the lecture time every week.
- ▶ Office: MAK D-2-240
- ▶ Office hour: 2:30 PM - 3:30 PM Tuesday, Thursday and after the class



In this course, we'll delve into the world of data mining, uncovering valuable insights from vast datasets. We will explore techniques to identify meaningful patterns, correlations, and trends, and apply them to both real-world and synthetic data. Topics covered include data preprocessing, dimensionality reduction, data visualization, predictive modeling, model evaluation, clustering, and association analysis techniques.

- ▶ Course Homepage:
 - ▶ Blackboard: <https://lms.gvsu.edu/>
- ▶ Prerequisites
 - ▶ A solid understanding of data structures and algorithms.
 - ▶ Basic knowledge in linear algebra and statistics.
 - ▶ Proficiency in at least one programming language, coupled with practical programming experience.

- ▶ Understand the fundamentals of data mining and learn basic algorithms.
- ▶ Know how to apply these algorithms effectively in real-world scenarios.
- ▶ Provide a foundational course for those interested in pursuing research in data mining.

- ▶ Recommended: [Data Mining Concepts and Techniques \(4th Edition\)](#) by Jiawei Han, Jian Pei, and Hanghang Tong. Publication Date: 2023. (free at [GVSU library](#))
- ▶ References
 - ▶ [Think Python: How to Think Like a Computer Scientist](#) by Allen B. Downey. (free)
 - ▶ [Python Data Science Handbook](#) by Jake VanderPlas. (free)
 - ▶ [Applied Machine Learning in Python](#) by Andreas C. Müller. (free)
 - ▶ [Data Mining: The Textbook](#) by Charu Aggarwal. (free)
 - ▶ Data Mining by Pang-Ning Tan, Michael Steinbach, and Vipin Kumar.
 - ▶ [Machine Learning](#) by Tom Mitchell. (free)
 - ▶ Introduction to Machine Learning by Ethem ALPAYDIN.
 - ▶ Pattern Classification by Richard O. Duda, Peter E. Hart, David G. Stork.
 - ▶ [The Elements of Statistical Learning: Data Mining, Inference, and Prediction](#) by Trevor Hastie, Robert Tibshirani, and Jerome Friedman.
 - ▶ Pattern Recognition and Machine Learning by Christopher M. Bishop.

- ▶ Quizzes & Homework assignments 30%
- ▶ Project 30%
- ▶ Midterm exam 20%
- ▶ Final exam 20%

Grade A	Grade B	Grade C	Grades D & F
$A \geq 93\%$	$B+ \geq 87\%$	$C+ \geq 77\%$	$D+ \geq 67\%$
$A- \geq 90\%$	$B \geq 83\%$	$C \geq 73\%$	$D \geq 60\%$
	$B- \geq 80\%$	$C- \geq 70\%$	$F < 60\%$

Homework: (individual work)

- ▶ Homework assignments are designed to reinforce course material and include a mix of programming tasks along with mathematical or written questions.
- ▶ Deadline: 11:59 pm Michigan time on the due date.
 - ▶ *Late policy*: Late assignment will not be accepted. Let me know if you need accommodation or have special reasons ASAP.
 - ▶ No copying or sharing of homework. But you can discuss general challenges and ideas with others

Course project: (Group work)

- ▶ **3-4 people for one group**
- ▶ **Goal: Solve a given data mining problem**
 - ▶ Kaggle Competition style. Choosing from the topics:
 - ▶ AI for K-12 MLL students STEM education
 - ▶ Financial market data (e.g. stocks, NFT or crypto currency price)
 - ▶ Detail will be announced later in the semester. Other topics will subject to instructors' approval
- ▶ **You are expected to submit a project report and demo your results at the end of the semester**

- ▶ Document all collaborations.
- ▶ No electronic code transfers between students.
- ▶ Code you find on the internet must be cited, with an active link to that code. That code should not solve the entirety of an assigned problem/project (i.e., don't have someone else do your project for you).
- ▶ You are encouraged to engage in conversations in online forums, but do not post solutions or solicit others to complete your work for you.
- ▶ You are encouraged to talk about problems with each other in non-technical terms (i.e., not code)
- ▶ Ultimately, you are responsible for all aspects of your submissions. You should be able to explain and defend your submission if the work is entirely your own.
 - ▶ **Suspicious cases will be reported to the Academic Honesty Committee**

Tentative Course Content

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Module	Lecture & Discussion Topic
1	Introduction and Know Your Data
2	Data Quality and Preprocessing
3	Dimensionality Reduction
5	Predictive modeling: regression
4	Predictive modeling: classification
6	Clustering
7	Neural Networks
8	Sequence analysis

Expectations for Success

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- ▶ Check Blackboard on a regular basis for announcements and assignments
- ▶ I'm here to help you! Come to office hours or schedule an appointment when you're feeling lost/stuck.
- ▶ Let me know when something is getting in the way of your success in this class.
- ▶ Let me know how the class and my teaching can be improved.
- ▶ Adhere to class, CIS, and GVSU policies on academic honesty.

GVSU provides opportunities for students to improve your academic skills through resources, such as

- ▶ Writing center (<https://www.gvsu.edu/wc/>)
- ▶ Speech lab (<https://www.gvsu.edu/speechlab/>)
- ▶ Research consultants (<https://www.gvsu.edu/csce/>)
- ▶ Library liaisons (<https://www.gvsu.edu/library/about-the-university-libraries-3.html>)
- ▶ Academic success center (<https://www.gvsu.edu/sasc>)

Warming Up

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Individual introduction (please post in blackboard)

- Name and what do you want to be called
- Describe your background and experience in Data Mining or Machine Learning.
- Specific topics you seek to learn from this class

data scientists job

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<https://www.amazon.jobs/content/en/job-categories/data-science>

<https://www.google.com/about/careers/applications/jobs/results?q=education%20scientists>

<https://openai.com/careers/data-scientist-product-san-francisco/>